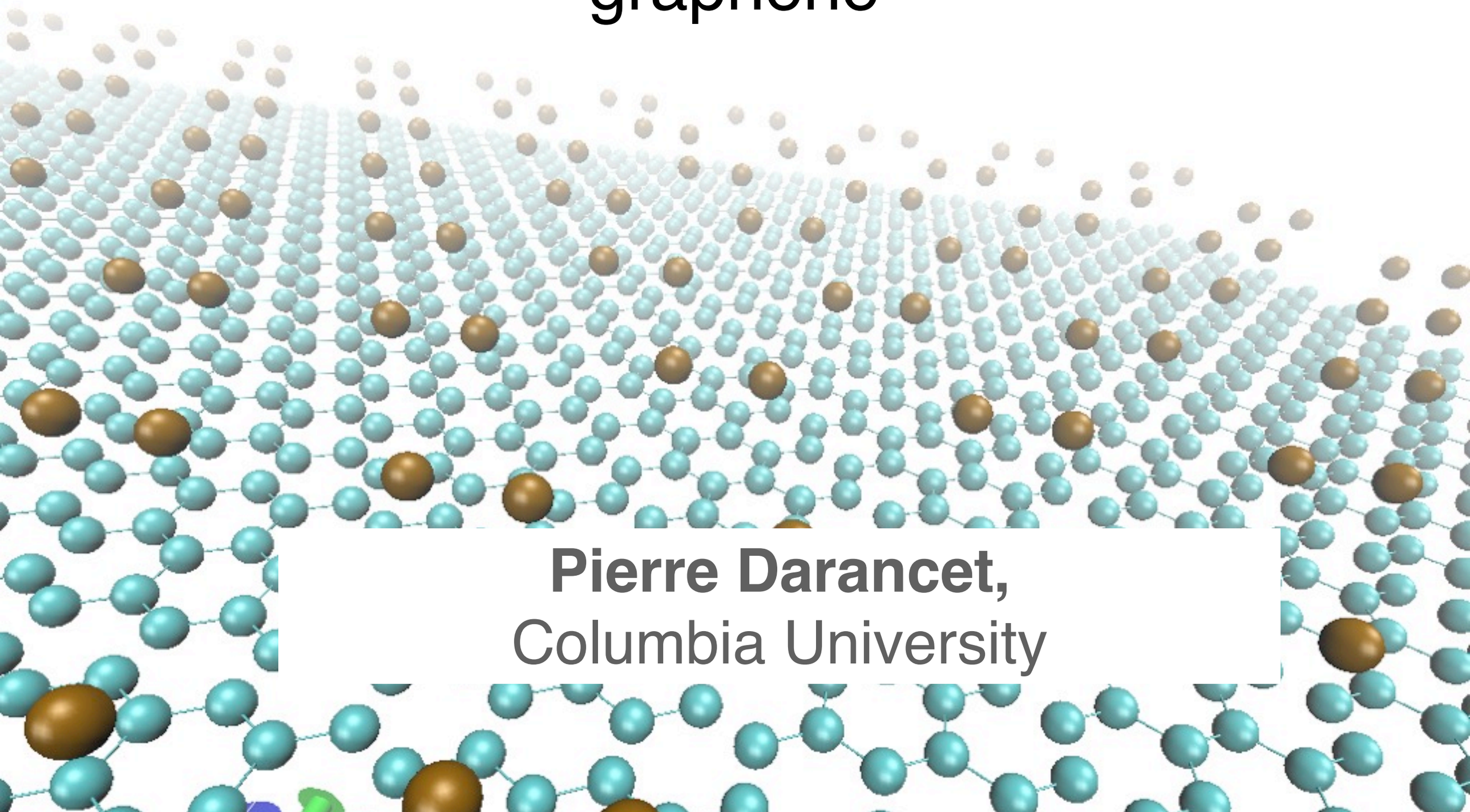
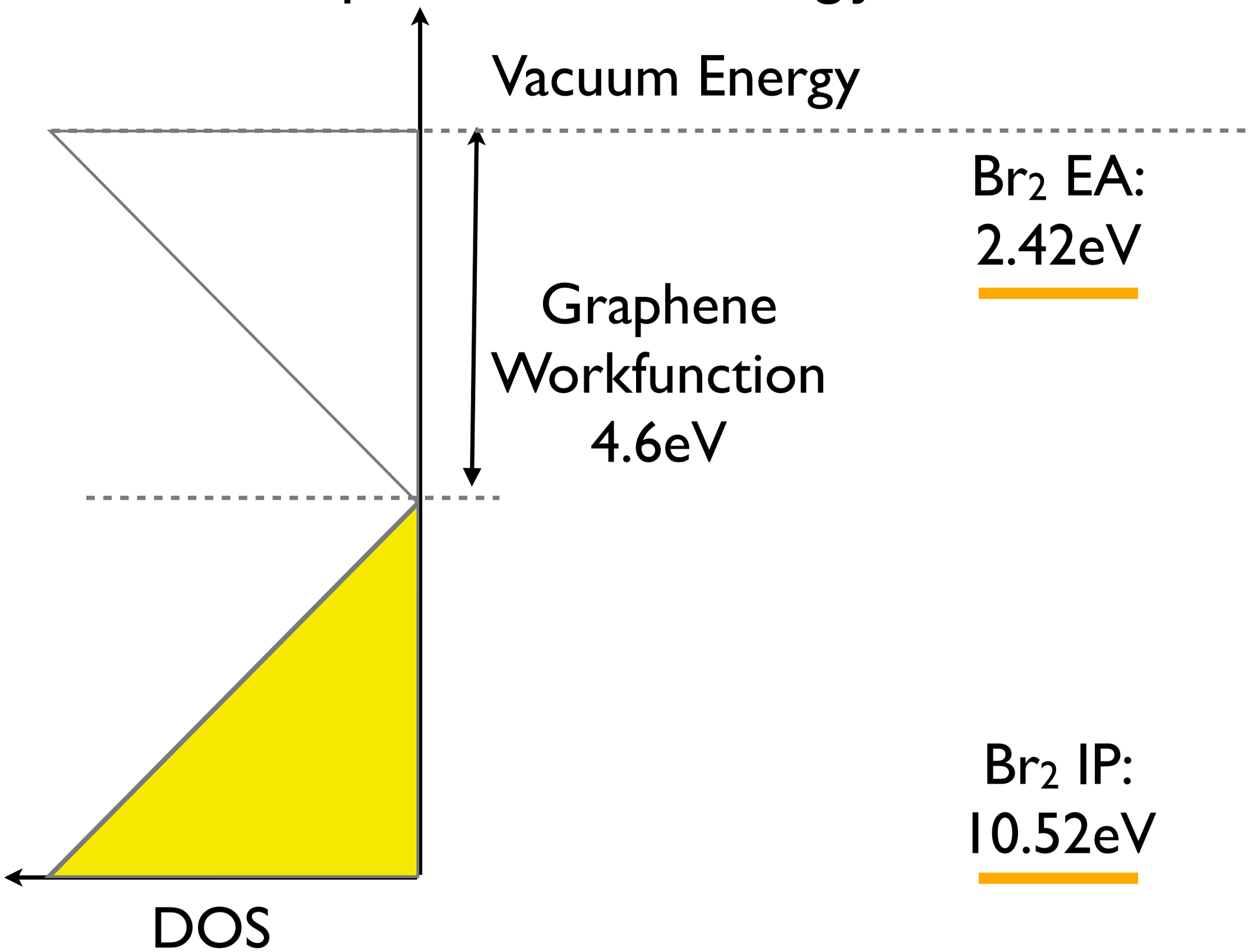
COLUMBIA | ENGINEERING
The Fu Foundation School of Engineering and Applied Science

Coverage-dependent Charge transfer in Br_2 on graphene

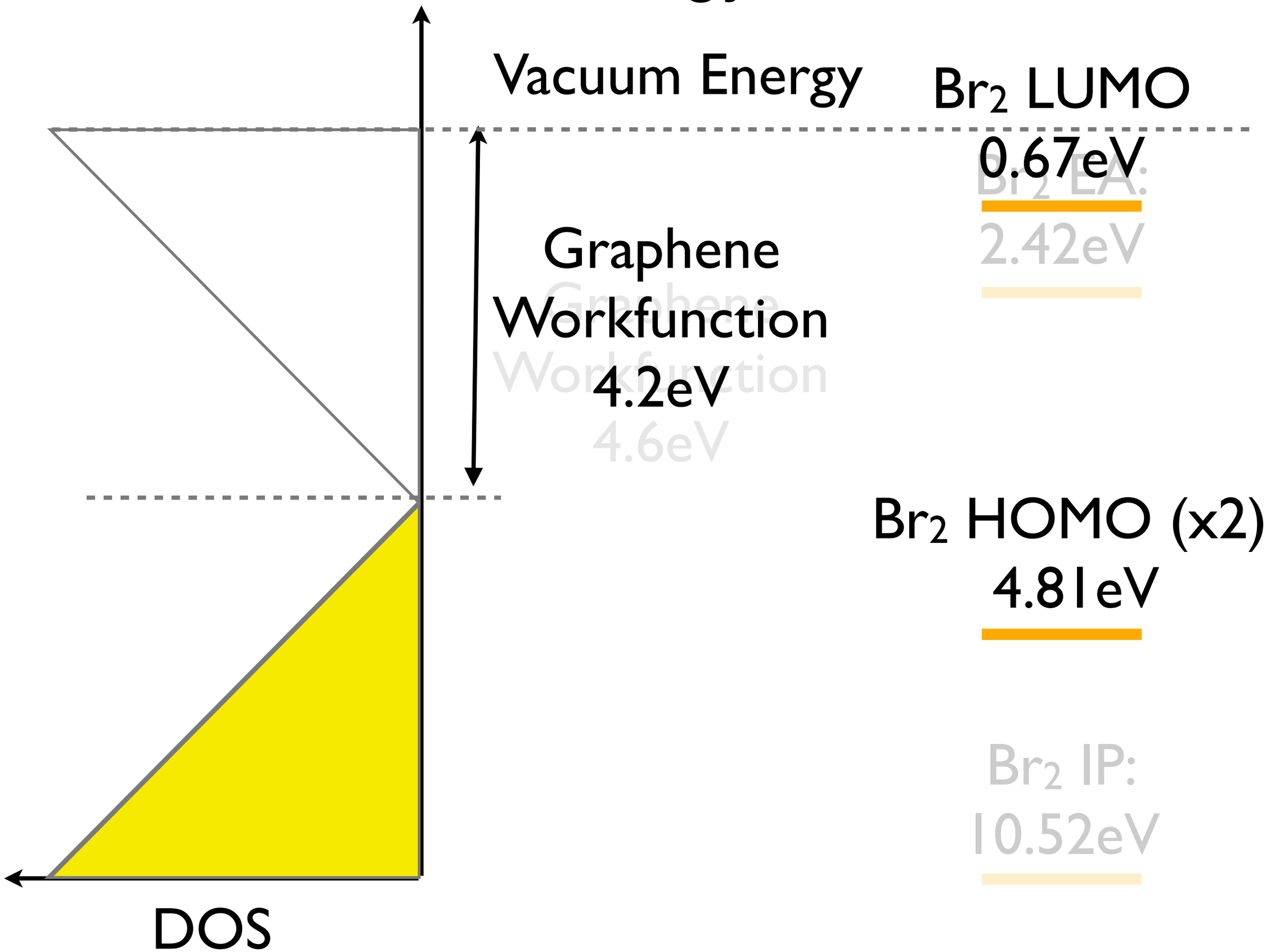


Pierre Darancet,
Columbia University

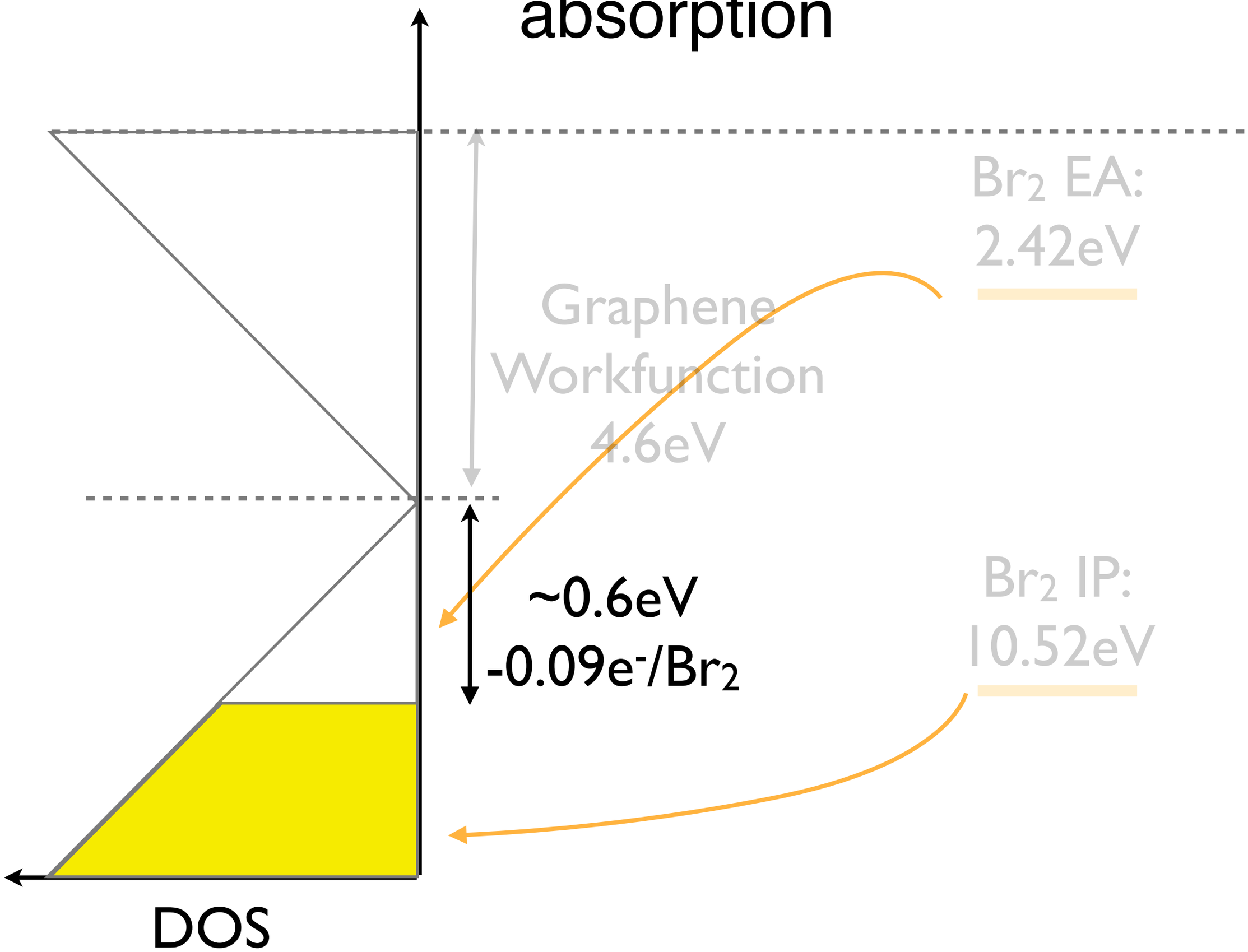
Experimental energy scales



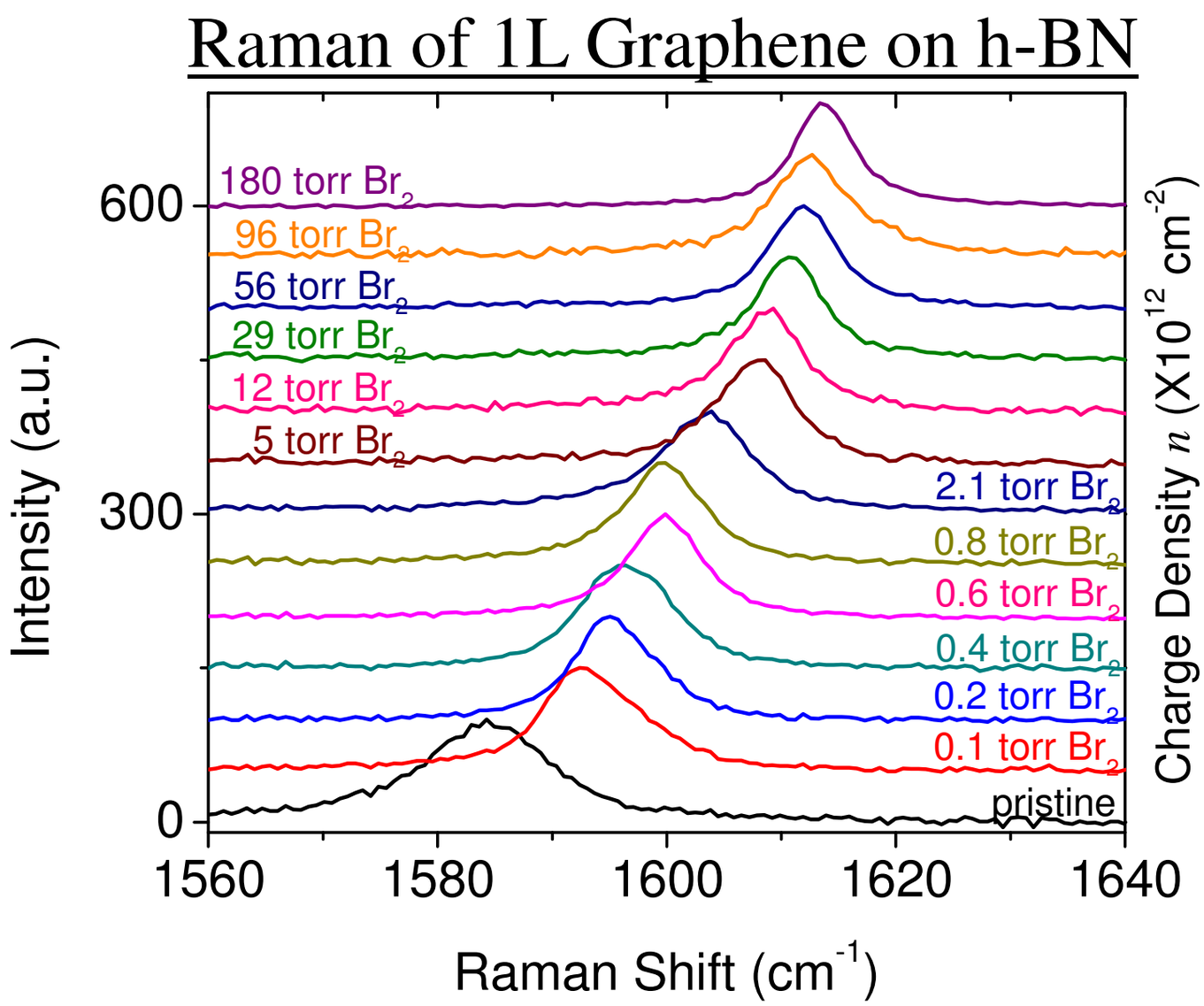
PBE energy scales

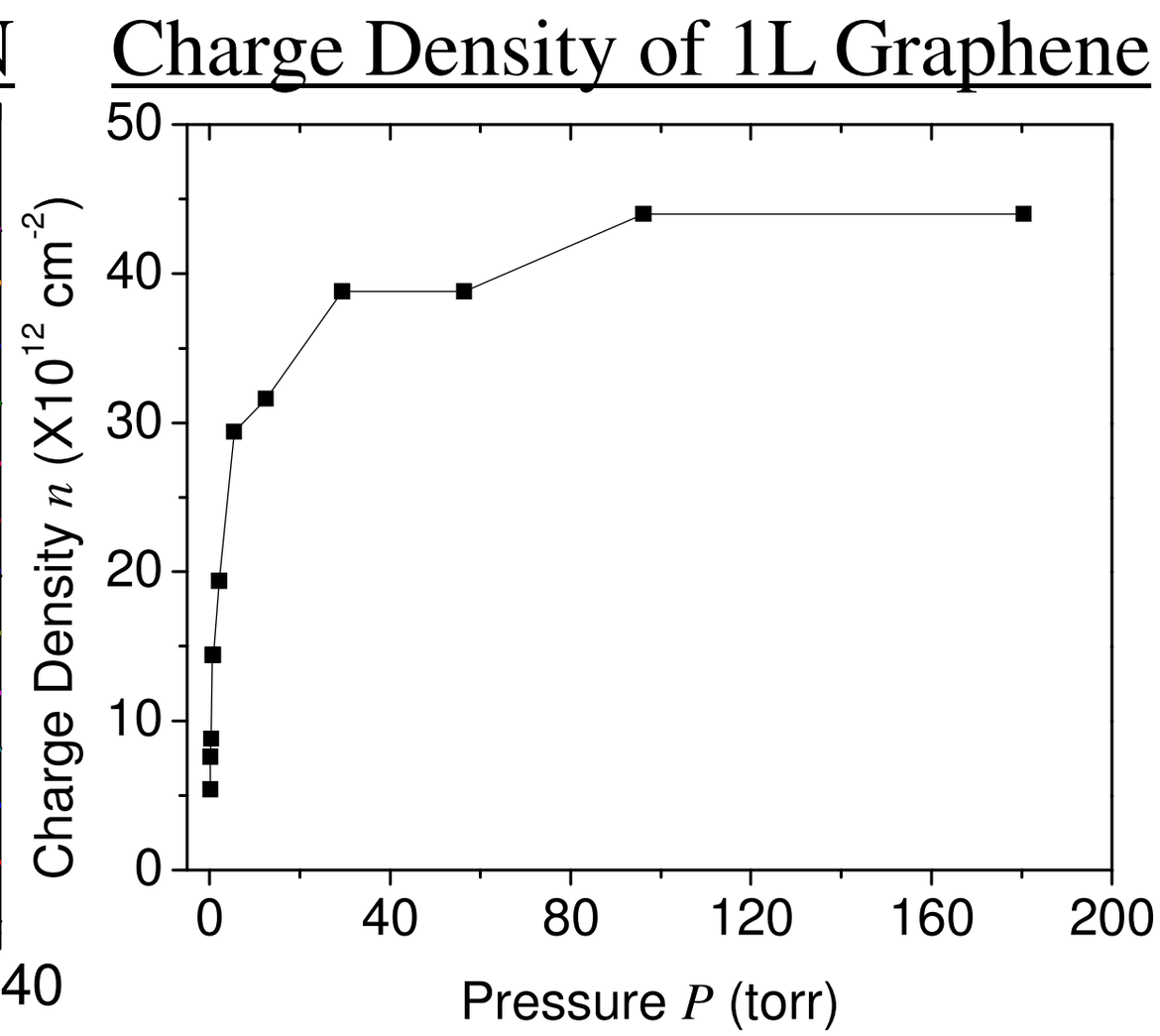


Experimental doping: Probed by Raman/Optical absorption



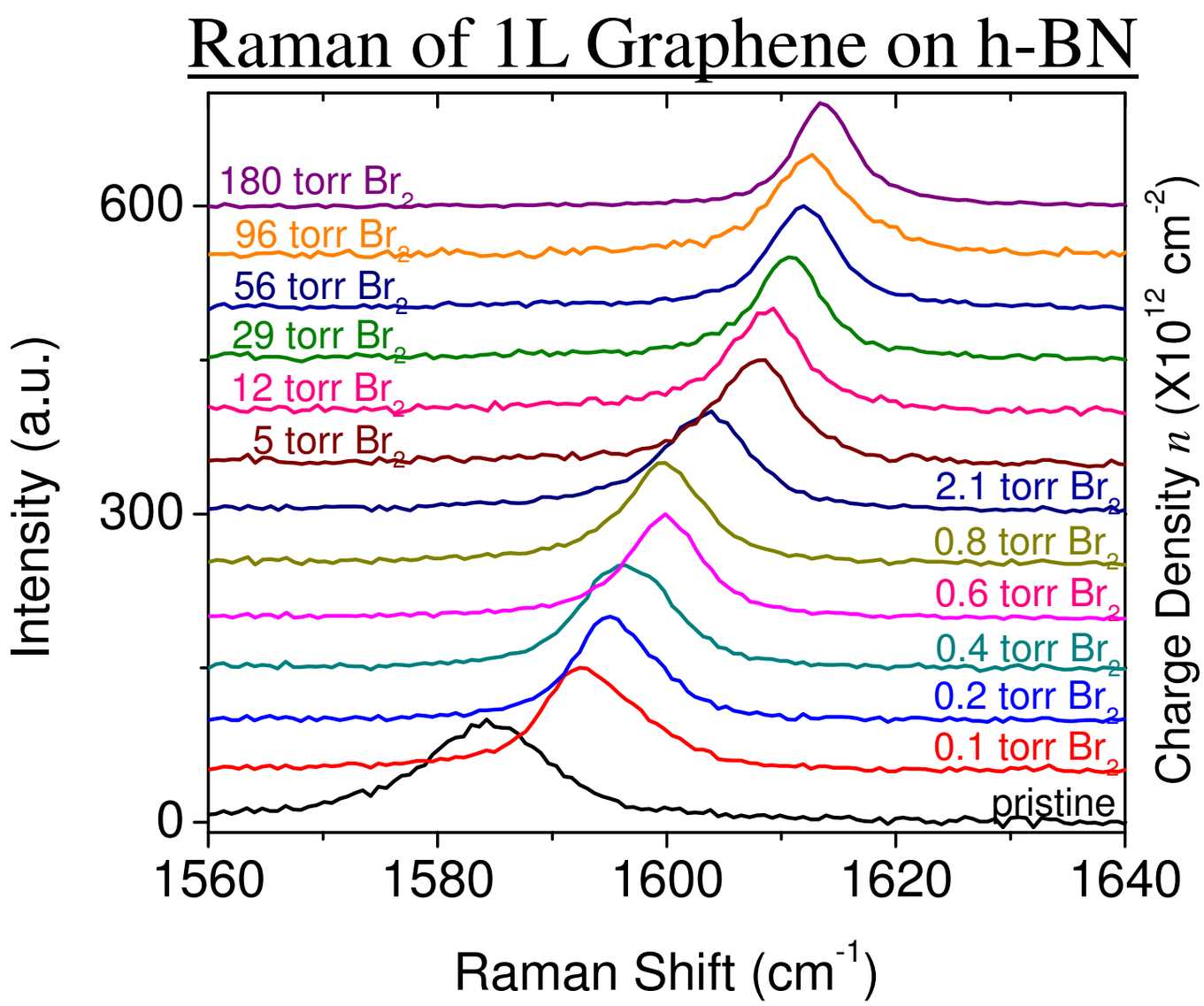
Br_2 Adsorbed on h-BN Supported 1L Graphene

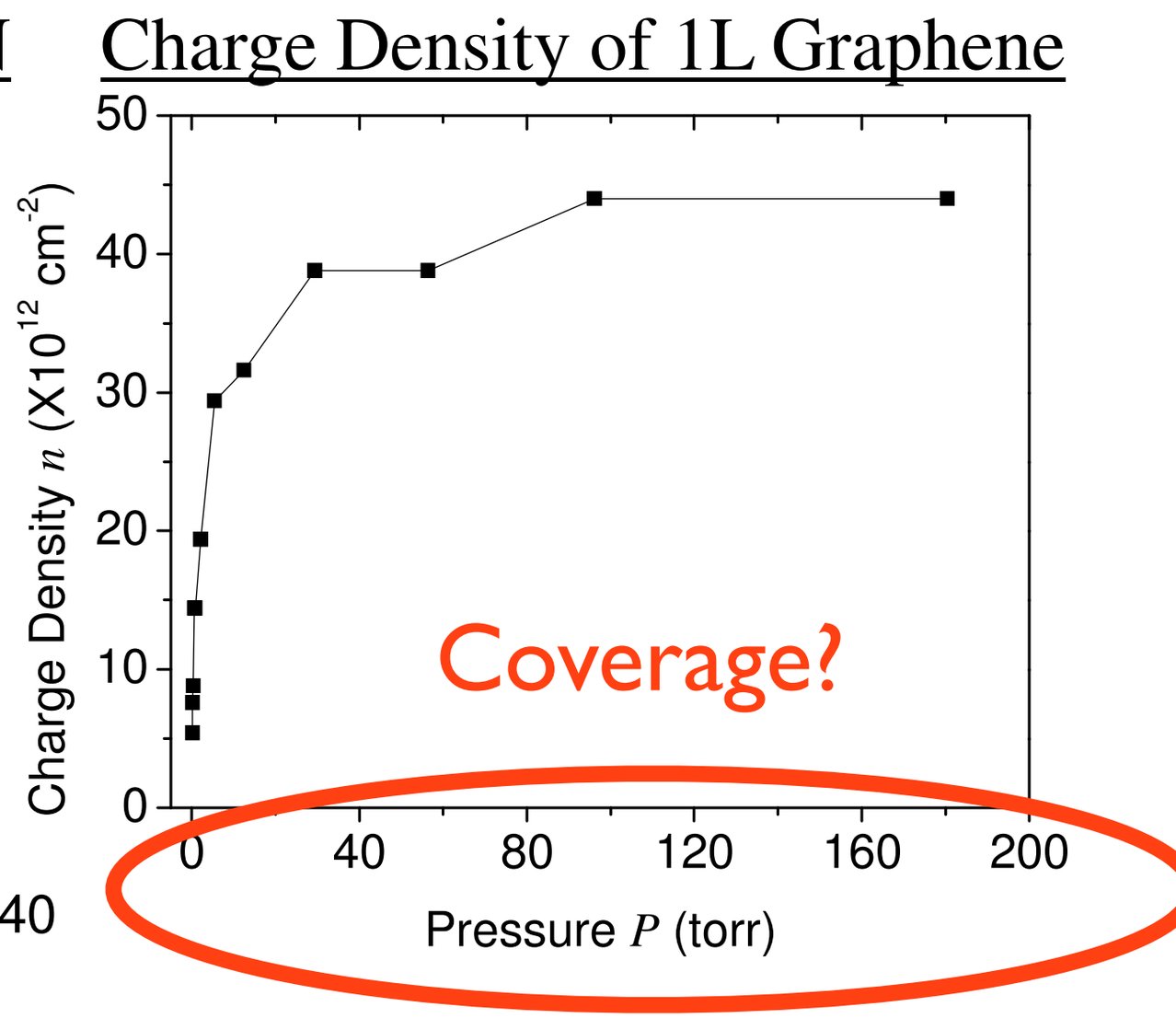




- Charge density *n* of 1L graphene is determined from the G peak position in Raman.
- Charge density increases with Br_2 pressure, and reaches the maximum of 47 x 10^{12} cm^{-2}.

Br_2 Adsorbed on h-BN Supported 1L Graphene





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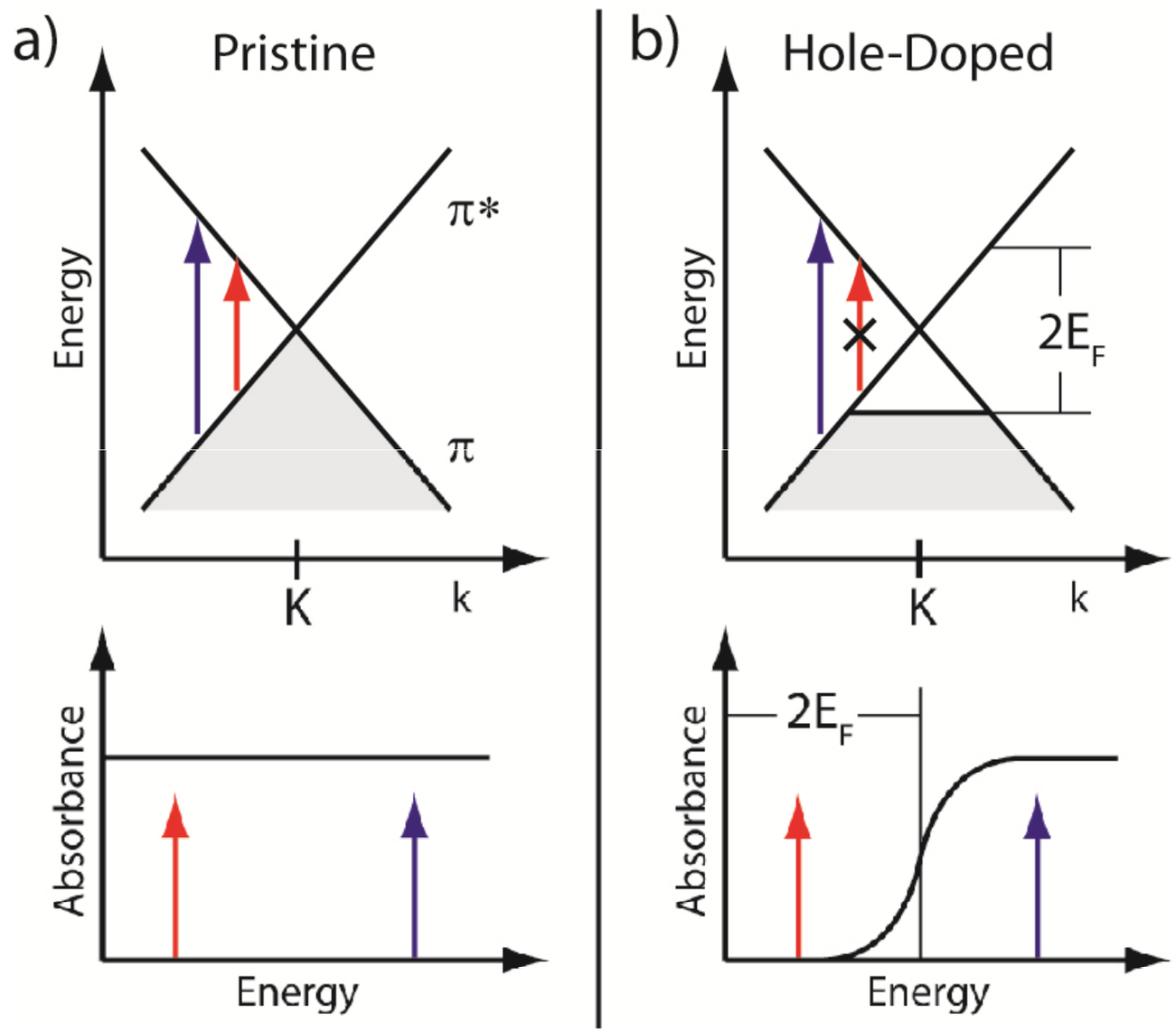
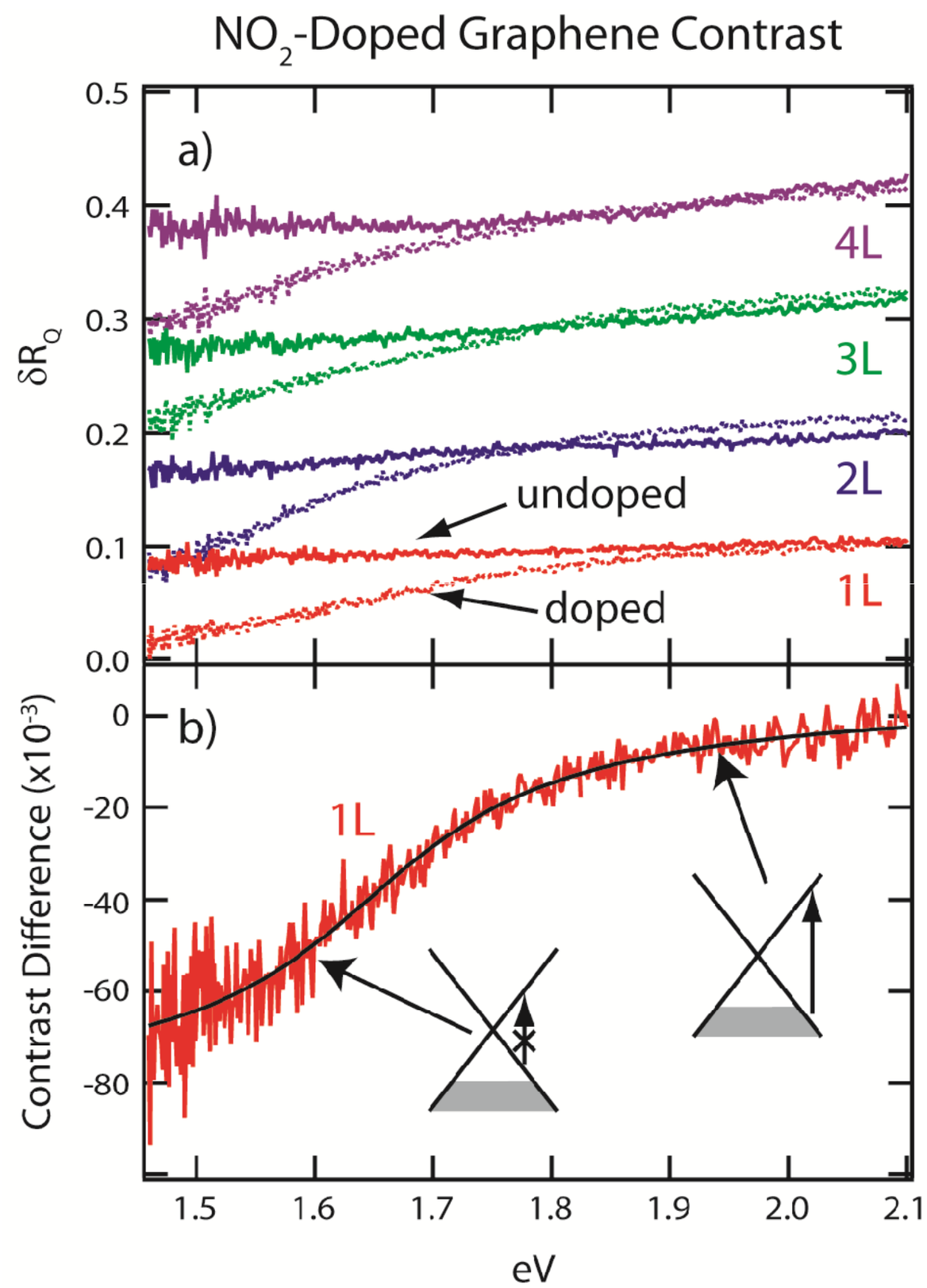
Bleaching of Optical Transition in Highly Doped Graphene

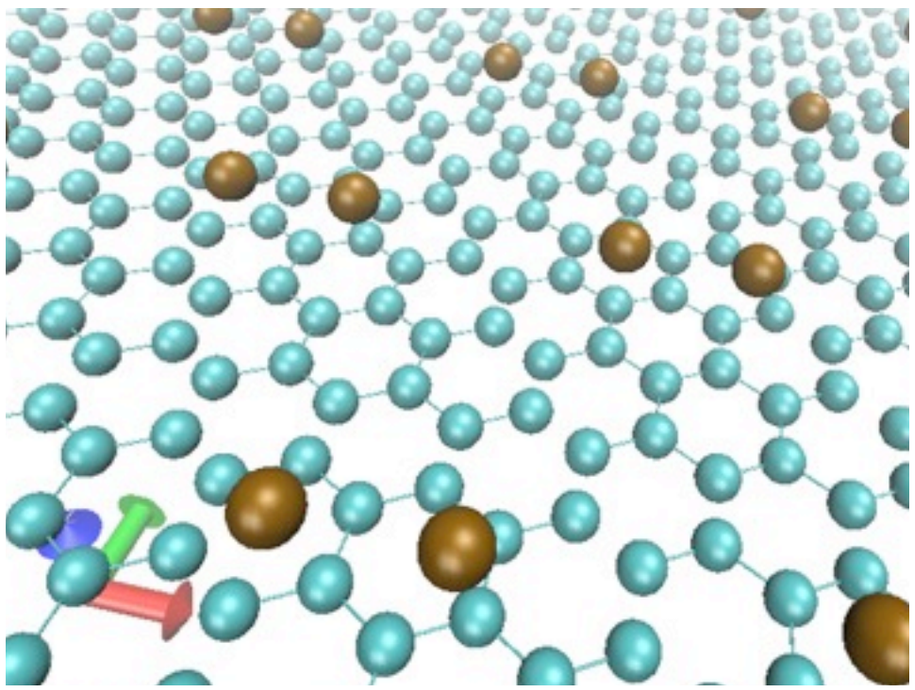




Gives similar results

A. C. Crowther *et al*, *ACS Nano* 2012

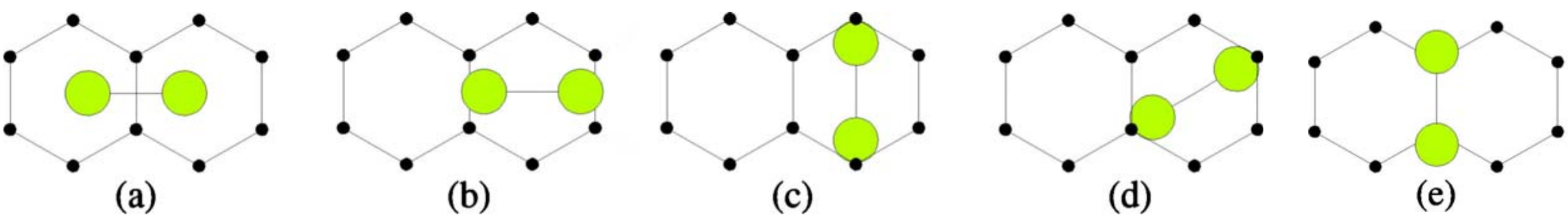
Simulated systems



Superstructure of Br_2 adsorbed on one side of graphene

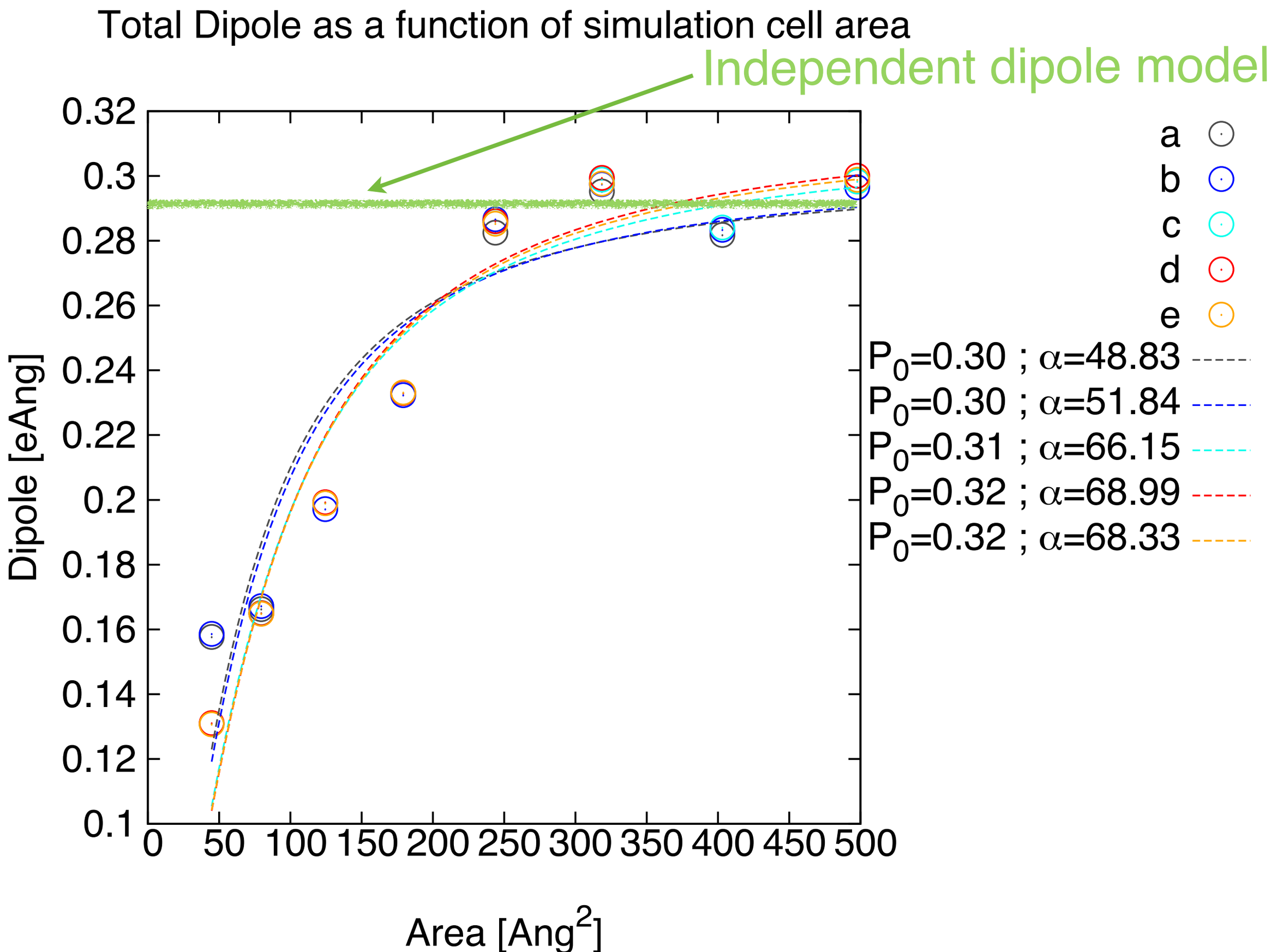
1 Br_2 per [8 ; 200] C-atoms

No structural relaxation



Br_2-C-plane distances: between 3.74 to 3.78Å

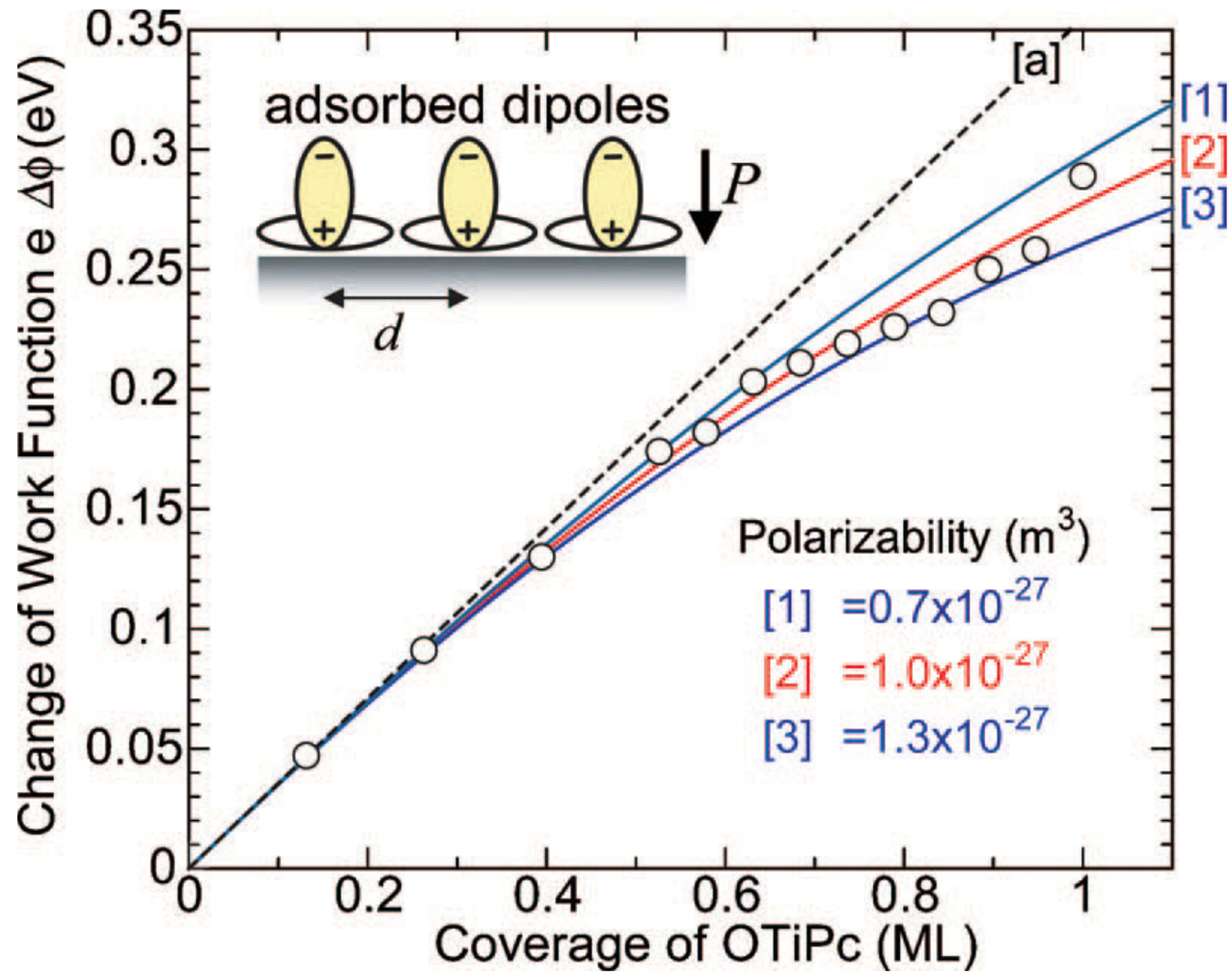
RUDENKO *et al.* PHYSICAL REVIEW B **82**, 035427 (2010)


Total Dipole as a function of simulation cell area
Independent dipole model
Dipole [eAng]
Area [Ang^2]
0.32
0.3
0.28
0.26
0.24
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0
50
100
150
200
250
300
350
400
450
500
a
b
c
d
e
P_0=0.30 ; α=48.83
P_0=0.30 ; α=51.84
P_0=0.31 ; α=66.15
P_0=0.32 ; α=68.99
P_0=0.32 ; α=68.33

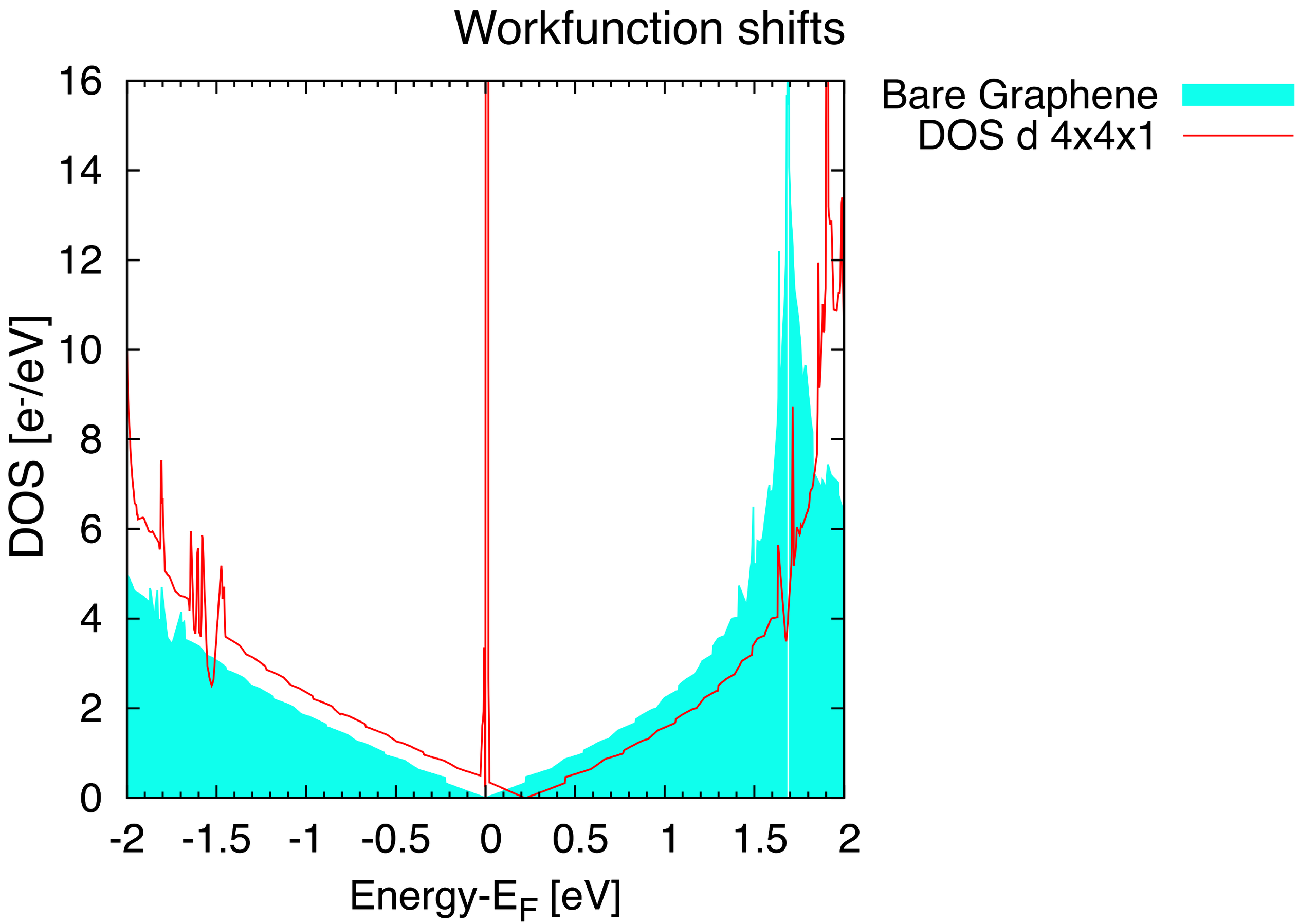
Polarizable dipole model

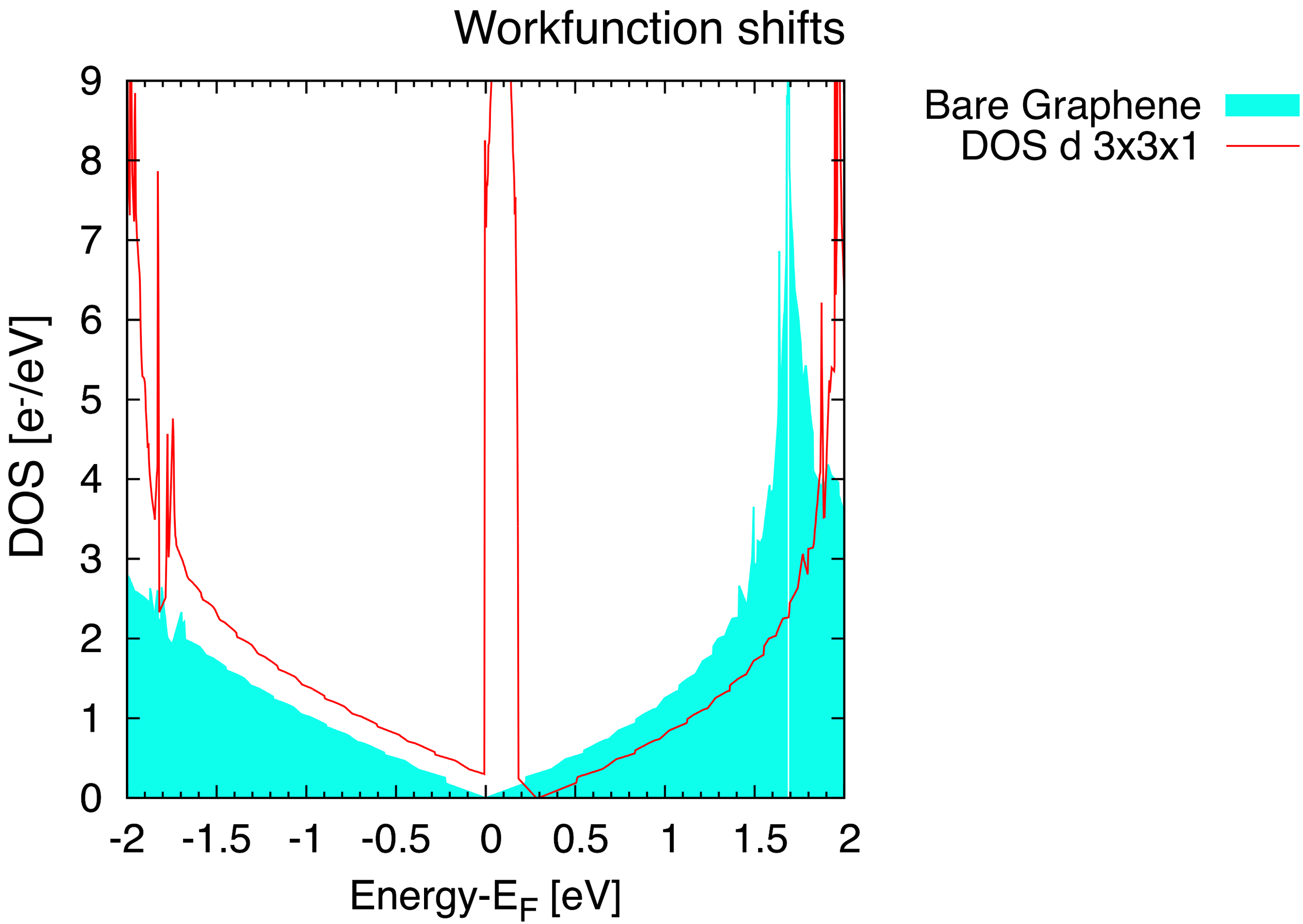
$$p = \frac{p_0}{1 + ak/a^3}$$

(a)

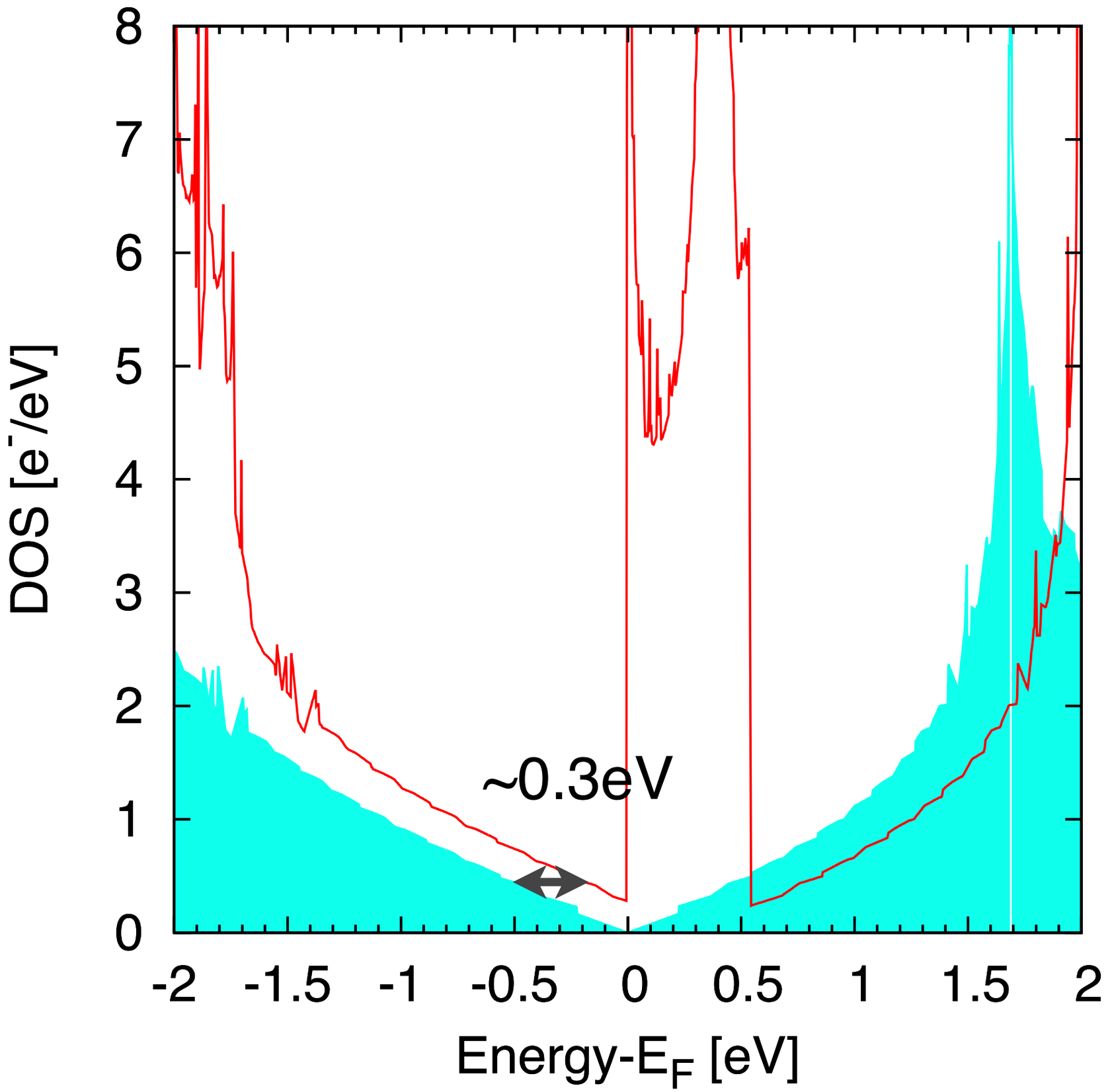


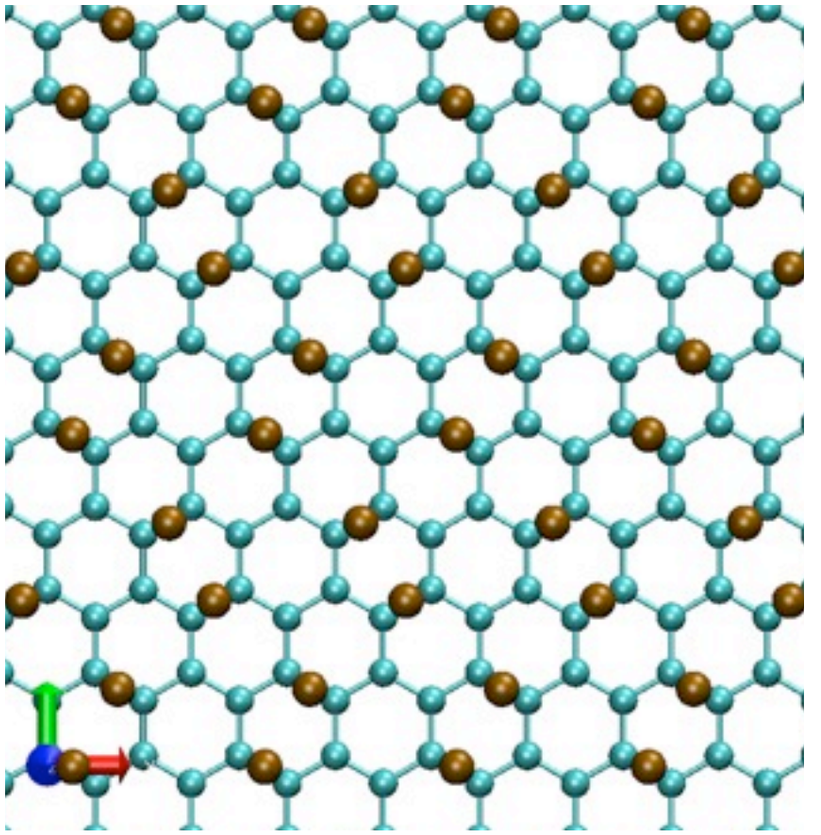
Amir Natan et al. ADVANCED MATERIALS


Workfunction shifts
Bare Graphene
DOS d 4x4x1
DOS [e-/eV]
Energy-E_F [eV]
16
14
12
10
8
6
4
2
0
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2

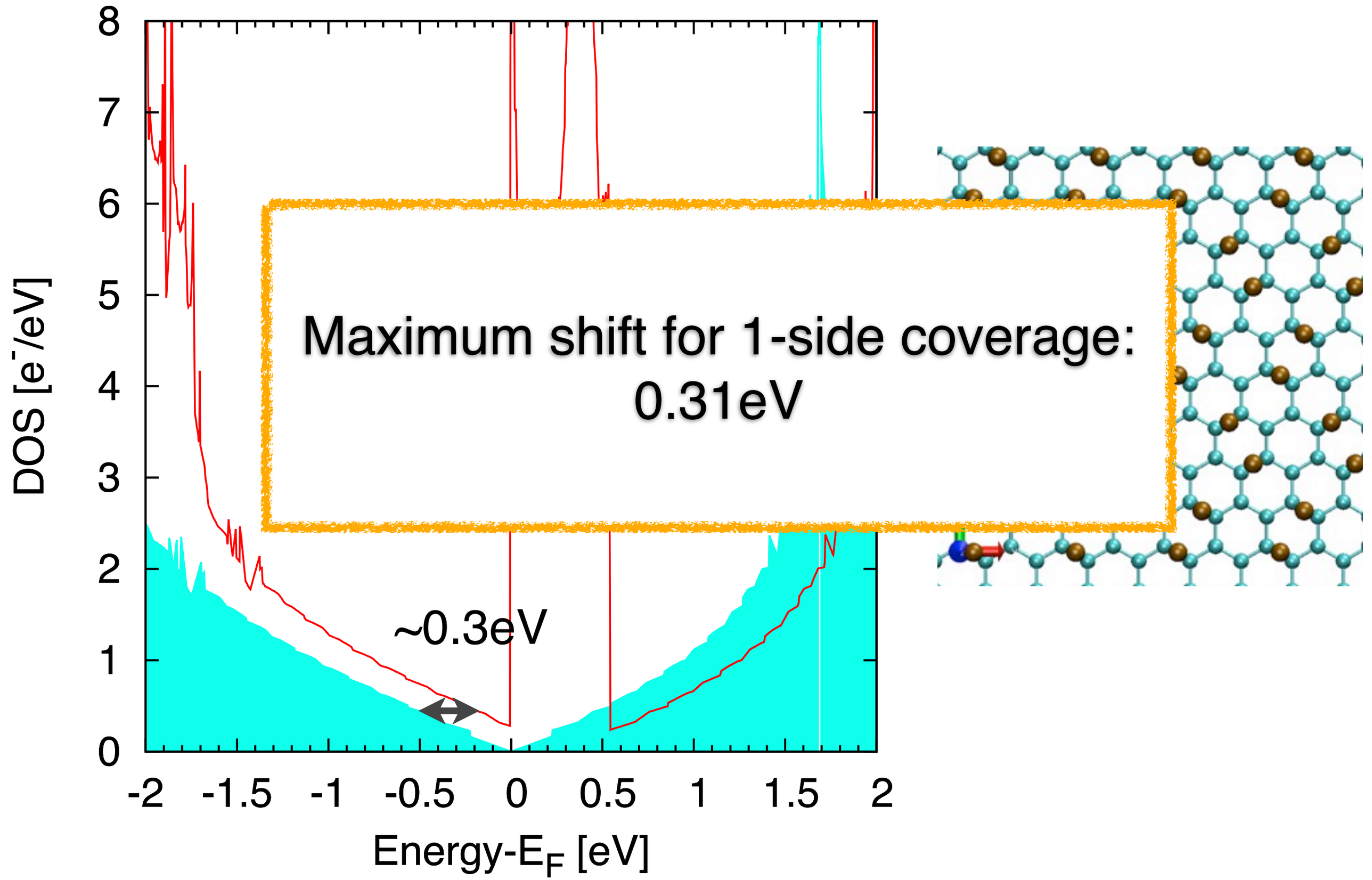

Workfunction shifts
Bare Graphene
DOS d 3x3x1
DOS [e-/eV]
Energy-E_F [eV]
9
8
7
6
5
4
3
2
1
0
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2

Workfunction shifts: Full Coverage

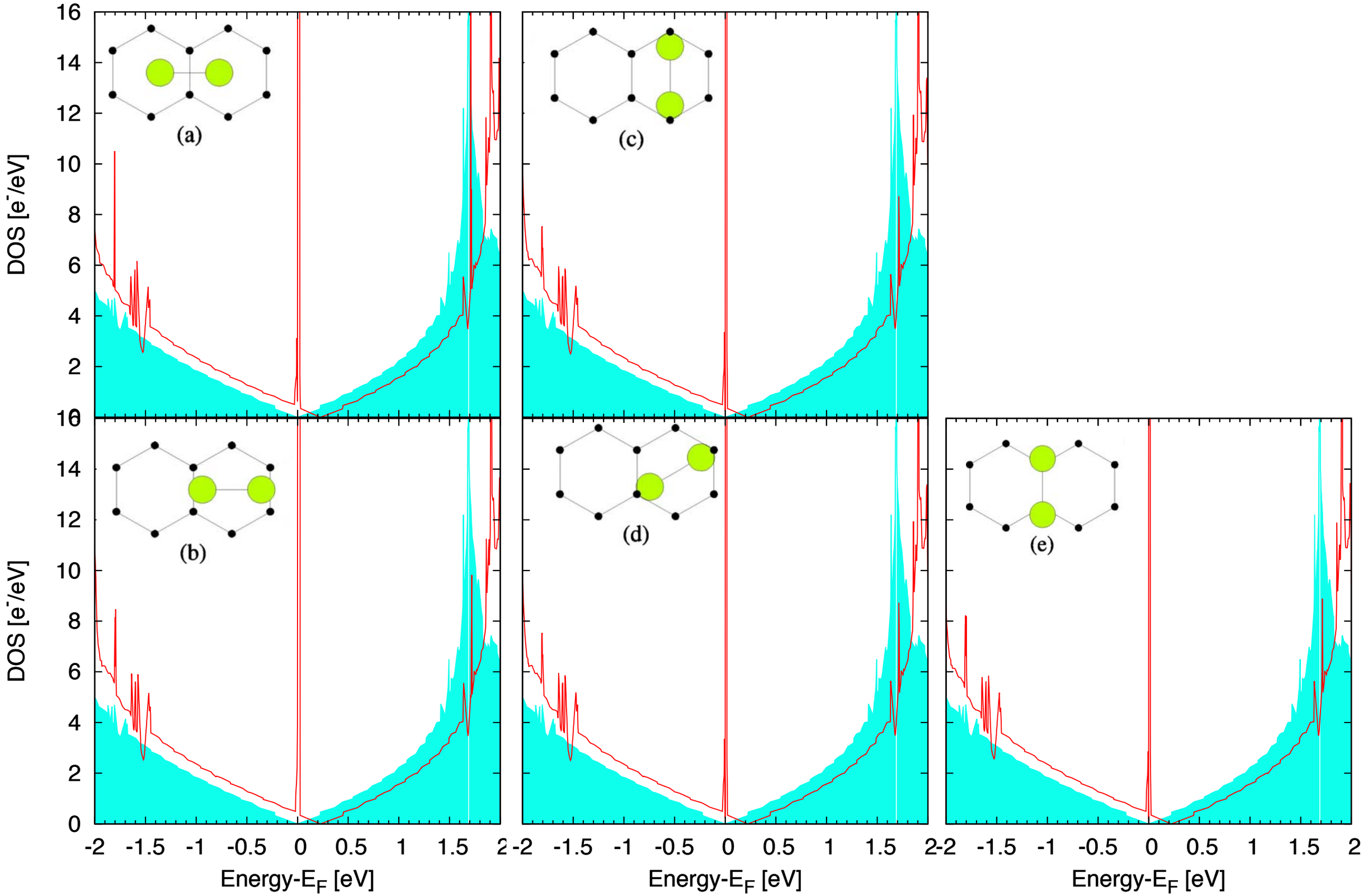




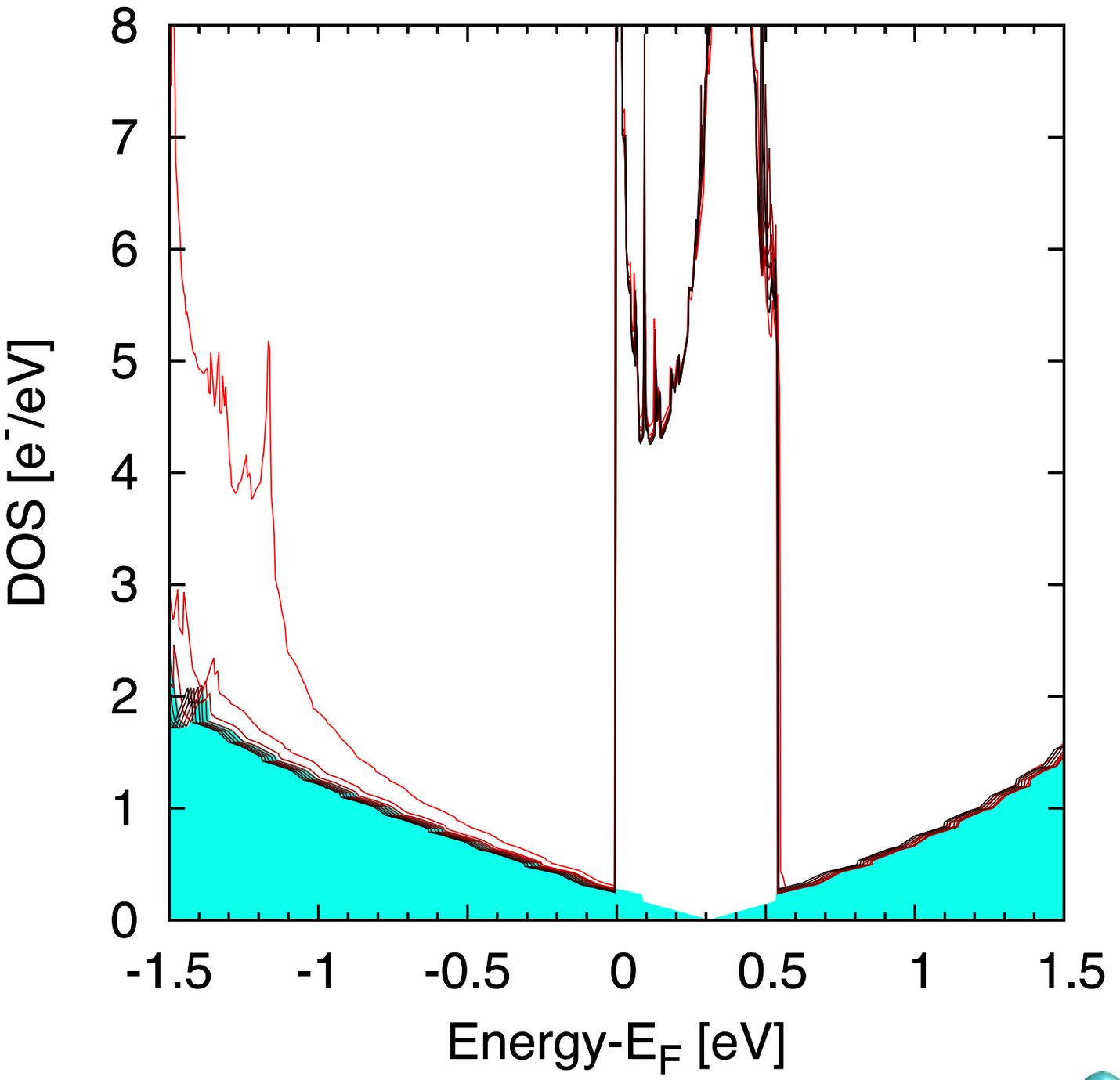
Workfunction shifts: Full Coverage

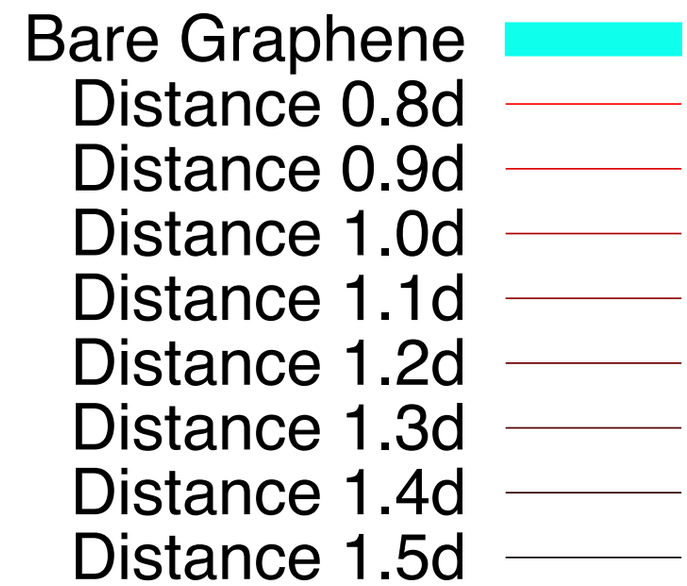


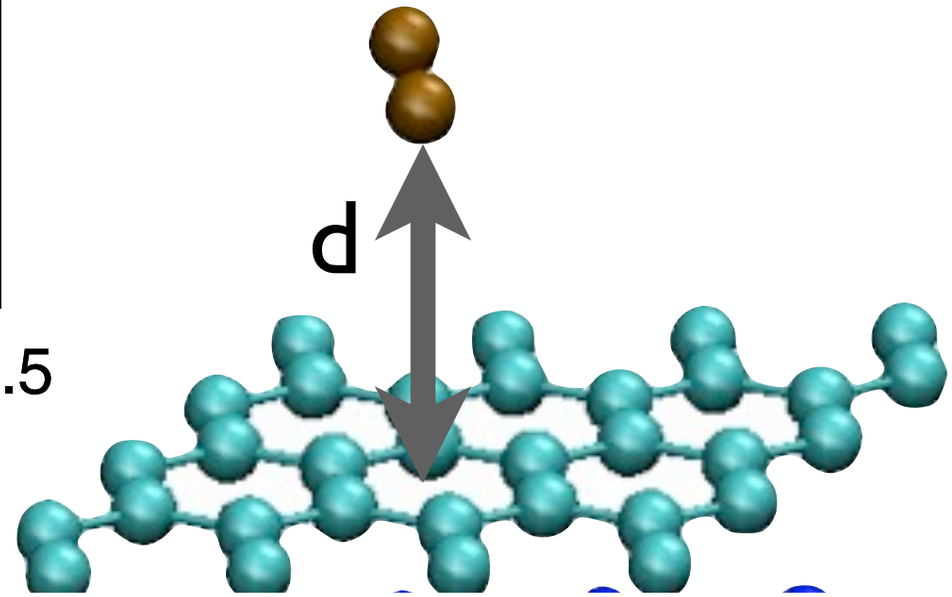
For coverages as low as Br2/32Carbon: No effect of the adsorption site



Effects of Br_2 vertical position







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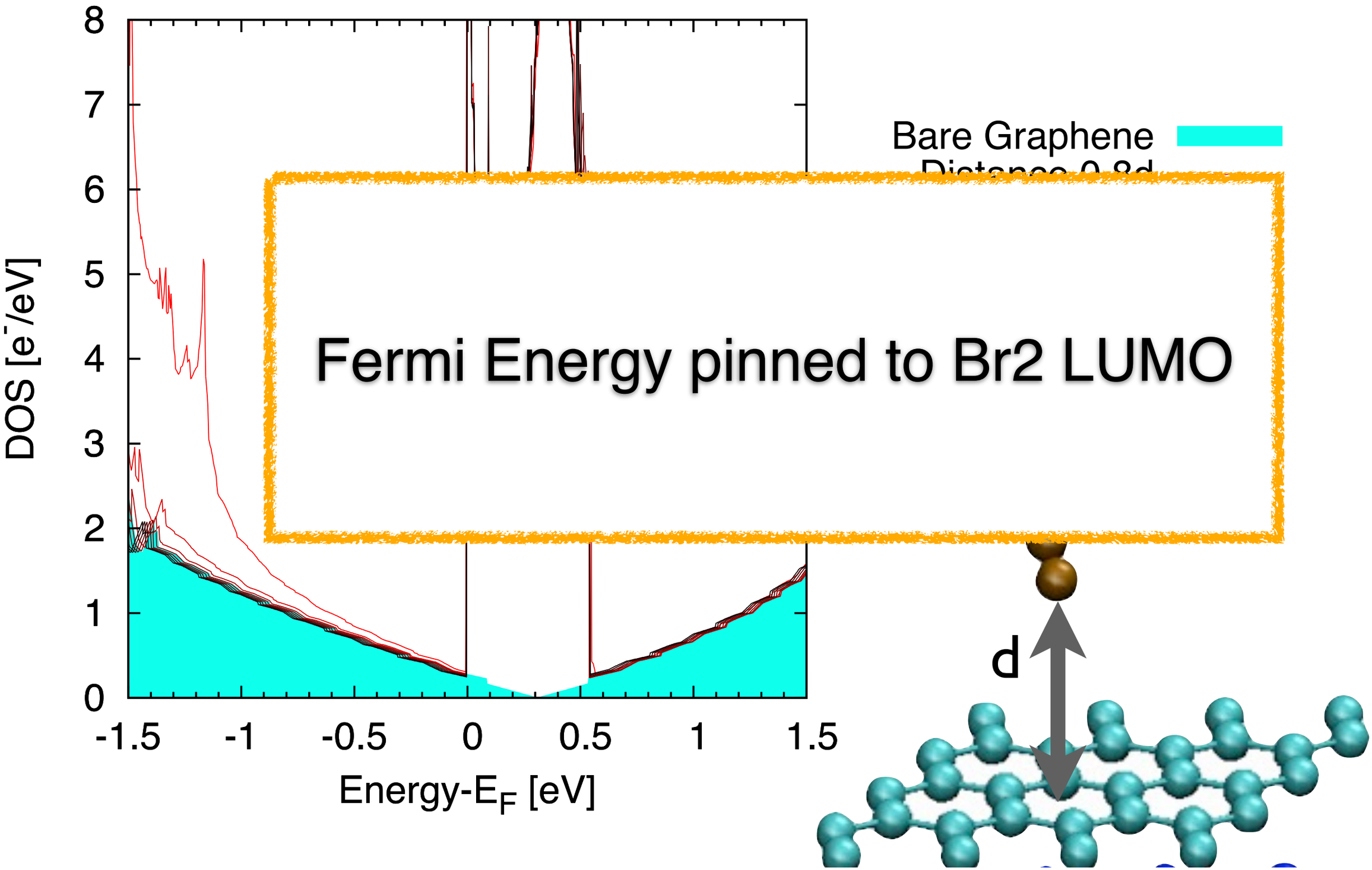
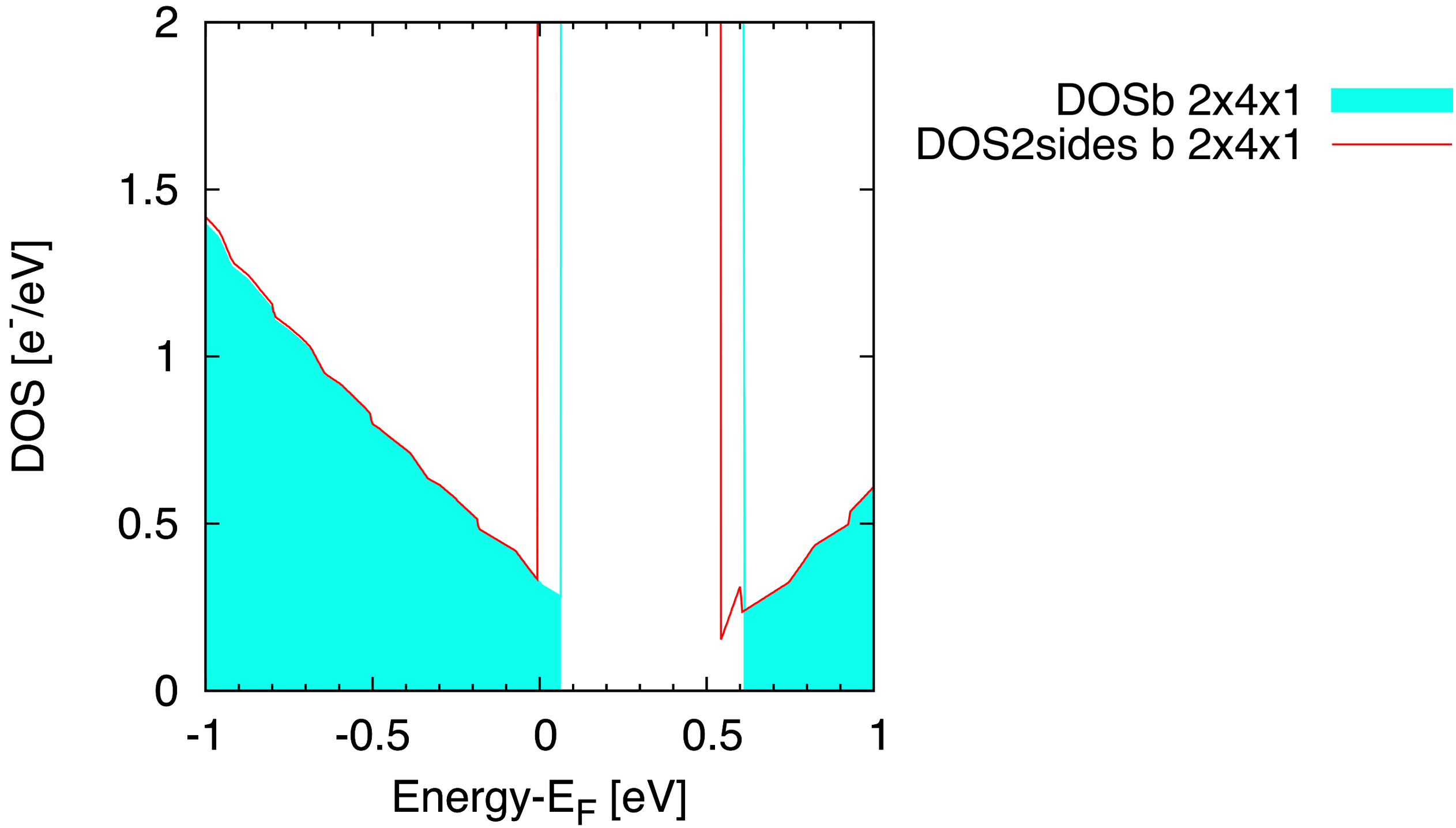
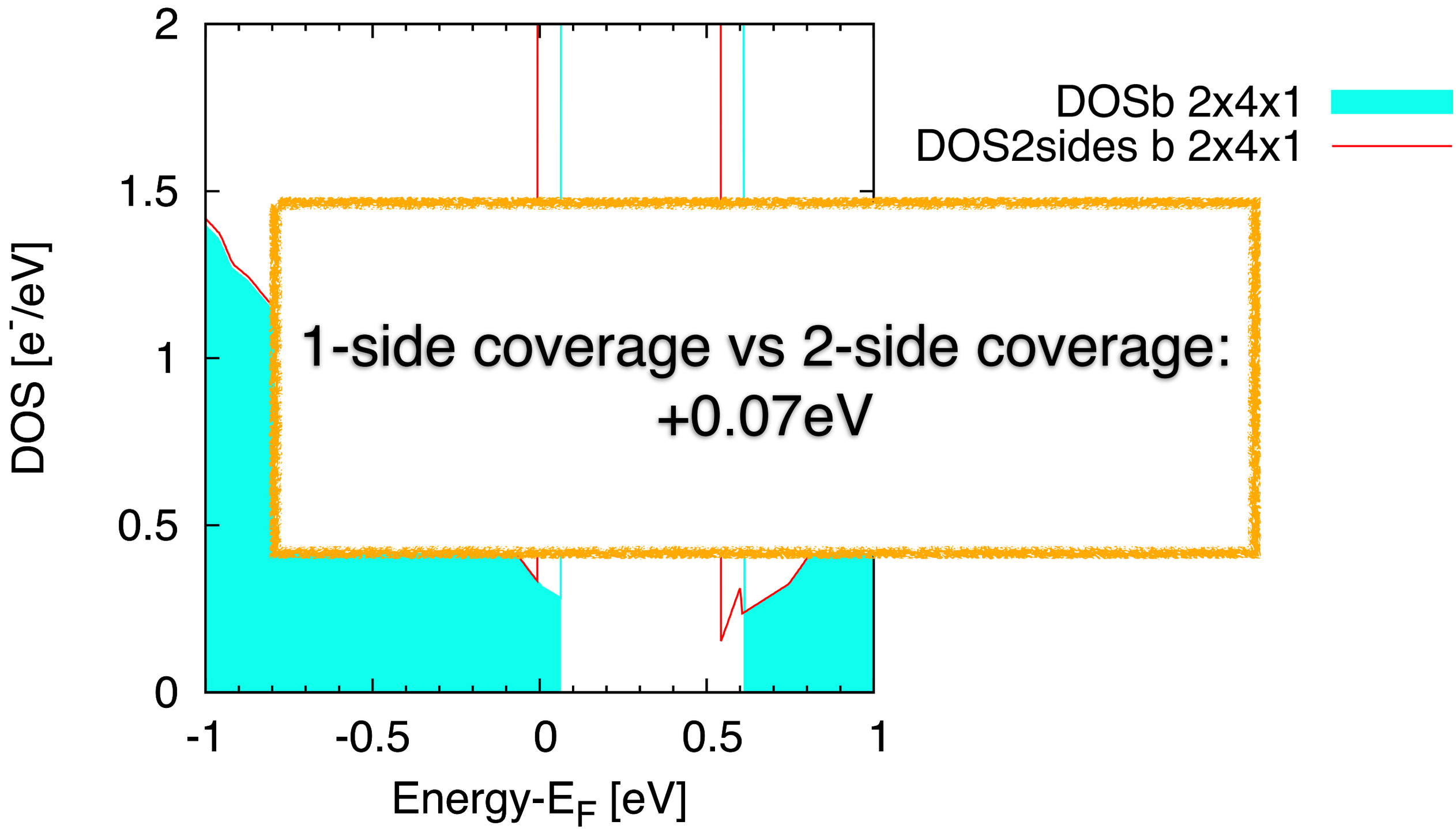


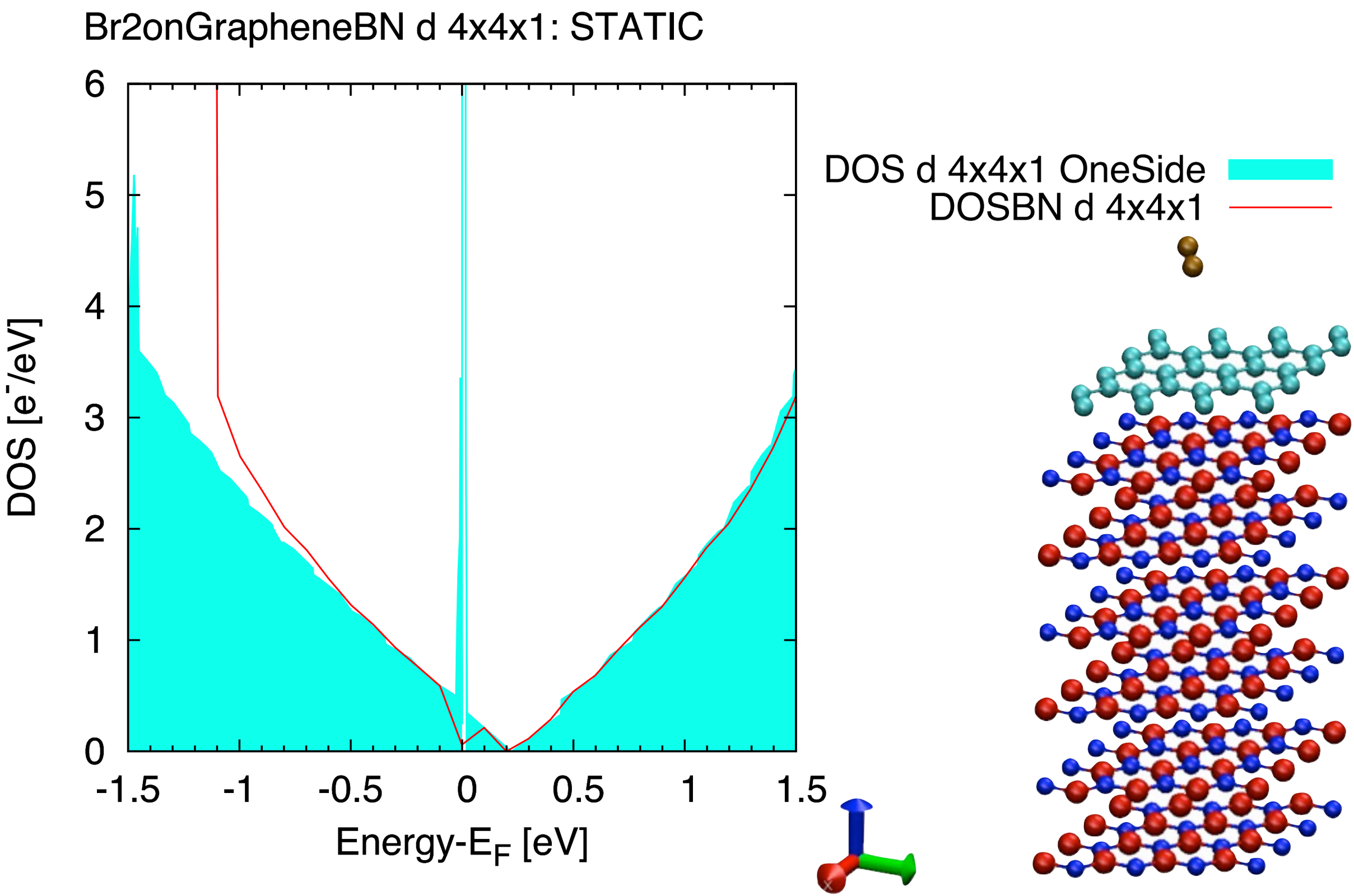
Two-sided coverage



Two-sided coverage



h-BN substrate



Conclusions

Maximum workfunction shifts obtained in DFT do not match experiment for Graphene@h-BN and 2-sided Graphene.

=> Due to pinning of the Fermi energy the Br2 LUMO.

Moving forward:

Better evaluation (MBPT or model correction) of LUMO energy

Better evaluation of the band-width at high coverages

Boltzman-average of the molecular positions

Other effects?